

Coding & Solution

Date	08 July 2026
Team ID	SL-2026-02
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/238x1a04i4-ux/Smart-Lender
Programming Language(s)	Python 3.x, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Bootstrap 5

Key Features Implemented	• Loan eligibility prediction using Machine Learning. • Applicant data preprocessing and feature engineering. • Multiple ML model training and evaluation. • Best model integration using Joblib/Pickle. • Flask-based web application for real-time prediction. • Responsive user interface using HTML, CSS, Bootstrap, and JavaScript. • Instant prediction results based on applicant information.
Pending / Incomplete Features	• User authentication and login system. • Cloud deployment (AWS/Azure/Render). • Loan application history and analytics dashboard. • Email/SMS notification service. • Database integration for storing prediction history.
Setup / Run Instructions	1. Install Python 3.x and required libraries using <code>pip install -r requirements.txt</code>. 2. Place the trained model (<code>model.pkl</code>) and preprocessing file (<code>transform.pkl</code>) in the project directory. 3. Open the project in Visual Studio Code or Jupyter Notebook. 4. Run <code>python app.py</code>. 5. Open the browser and navigate to http://127.0.0.1:5000/. 6. Enter applicant details and click Predict to view the loan eligibility result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The project follows a modular architecture, separating preprocessing, model training, prediction, and web application logic.
- The machine learning model is trained using historical loan applicant data and integrated into a Flask web application for real-time predictions.
- The solution emphasizes accuracy, maintainability, scalability, and ease of use.
- The application provides a responsive interface and supports quick loan eligibility prediction, reducing manual effort in the loan approval process.